Yu (Angel) Chu

Ph.D. Student, Mechanical Engineering | Graduate Research Assistant, NCSU
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Research Profile

Biomedical ultrasound researcher focused on wearable ultrasonic sensing, ultrasound transducer design, non-invasive cardiovascular and hemodynamic monitoring, through-tissue ultrasonic power transfer, and piezoelectric materials for sensing and imaging systems. Google Scholar profile reports 74 citations, h-index 5, and i10-index 2 as of June 9, 2026.

Education

North Carolina State University — Raleigh, NC
Ph.D. Student, Mechanical Engineering — May 2024–Present

- Research focus: biomedical ultrasound, wearable ultrasonic sensing, transducer design, non-invasive cardiovascular monitoring, and ultrasound-based physiological signal measurement.

Carnegie Mellon University — Pittsburgh, PA
M.S., Electrical and Computer Engineering – Applied Advanced Study — May 14, 2023

University of California, Santa Barbara — Santa Barbara, CA
B.S., Electrical Engineering — June 11, 2021

- UCSB Technology Management Program Certificate.

Research Experience

Micro/Nano Engineering Lab (MNEL), North Carolina State University — Raleigh, NC
Research Assistant — July 2023–Present

- Conduct biomedical-ultrasound research across wearable ultrasound sensing, through-tissue ultrasonic power transfer, blood-pressure and hemodynamic monitoring, musculoskeletal monitoring, neuromodulation-related systems, and intravascular-ultrasound-related technologies.
- Develop and evaluate wearable ultrasound platforms for continuous, non-invasive cardiovascular monitoring, including transducer integration, pulse-echo signal acquisition, physiological waveform analysis, and experimental validation.
- Design, fabricate, characterize, and compare ultrasound transducers using KLM modeling, COMSOL simulations, benchtop testing, phantom studies, ex vivo experiments, and in vivo or human-subject sensing workflows.
- Contribute to wearable and implantable ultrasound system development, including acoustic matching, materials selection, signal processing, experimental troubleshooting, and performance assessment.
- Prepare manuscripts, conference abstracts, figures, presentations, and technical documentation for journal submissions, conference dissemination, and internal research reporting.

Human Machine Interaction Lab, Tsinghua University — Beijing, China
Mechanical Engineering Intern — June 2017–September 2017

- Optimized hardware for haptics and force-feedback gloves using SolidWorks, AutoCAD, Adams, and 3D printing.
- Generated test procedures to assess constraint-space performance and support functional validation of glove prototypes.

Industry Experience

Google — Sunnyvale, CA
Data Center Engineer Intern — May 2022–August 2022

- Performed SQL- and Python-based analysis of data-center low-voltage bus-level loads, distribution, and utilization.
- Created an online simulator for testing and reviewing interactions between SSS SWGR and RMU loops.
- Translated relay logic for next-generation data-center electrical-system architecture from Siemens DIGSI relay settings into PLC logic in CoDeSys, verified logic in simulation scenarios, and built a GUI for single-line-diagram operation in a PLC environment.

Luxshare Precision Industry Co. Ltd. — San Diego, CA
AI Research and Development Intern — June 2021–September 2021

- Conducted VR/AR smart-glasses market analysis and summarized product opportunities for management decision-making.

Peer-Reviewed Journal Articles and Reviews

1. **Yu Chu**, A. Naderi, H. Wu, X. Xue, M. L. Oelze, and X. Jiang, "Recent Advances in Transducers for Through-Tissue Ultrasonic Power Transfer," *Progress in Biomedical Engineering*, 2026. doi:10.1088/2516-1091/ae5f8a.
2. H. Wu, H. Wan, **Yu Chu**, C. Luo, W.-Y. Chang, M. Chen, and X. Jiang, "Study on Ultrasound Transducers with Alternating Current Poled $Pb(Mg_{1/3}Nb_{2/3})O_3$-$x$$PbTiO_3$ Single Crystals," *IEEE Transactions on Ultrasonics*, 2026, pp. 1–1. doi:10.1109/TUSON.2026.3657788.
3. X. Xue, S. Moon, V. Ganesh, Q. Cai, **Yu Chu**, E. Schropp, D. Roque, T. P. Garcia, N. Sharma, and X. Jiang, "Wearable Ultrasound Sensing with Dual Arrays and Machine Learning for Real-Time Tremor Characterization and Antagonist Muscle Monitoring," *IEEE Transactions on Instrumentation and Measurement*, vol. 75, article 4001515, 2026. doi:10.1109/TIM.2026.3654713.
4. M. Chen, Z. Sheng, R. Wei, B. Zhang, H. Kim, H. Wu, **Yu Chu**, Q. Chen, A. Breon, S. Li, M. B. Wielgat, D. Shanmuganayagam, E. Tzeng, X. Geng, K. Kim, and X. Jiang, "Millisecond-Level Transient Heating and Temperature Monitoring Technique for Ultrasound-Induced Thermal Strain Imaging," *Theranostics*, vol. 15, no. 3, pp. 815–827, 2025. doi:10.7150/thno.95997.
5. T. Gao, Q. Liao, W. Si, **Yu Chu**, H. Dong, Y. Li, Y. Liao, and L. Qin, "From Fundamentals to Future Challenges for Flexible Piezoelectric Actuators," *Cell Reports Physical Science*, vol. 5, no. 2, article 101789, 2024. doi:10.1016/j.xcrp.2024.101789.
6. W. Si, Q. Liao, X. Wang, Z. Wang, **Yu Chu**, M. Sun, Z. Ran, X. Chu, and L. Qin, "Organic-Inorganic Piezoelectric Single-Crystal TMCM-2$CdCl_3$ with High Piezoelectric Properties," *Chemical Physics Letters*, vol. 835, article 141000, 2024. doi:10.1016/j.cplett.2023.141000.
7. Y. Liao, H. Yang, Q. Liao, W. Si, **Yu Chu**, X. Chu, and L. Qin, "A Review of Flexible Acceleration Sensors Based on Piezoelectric Materials: Performance Characterization, Parametric Analysis, Frontier Technologies, and Applications," *Coatings*, vol. 13, no. 7, article 1252, 2023. doi:10.3390/coatings13071252.
8. W. Si, Q. Liao, **Yu Chu**, Z. Zhang, X. Chu, and L. Qin, "A Multi-Layer Core–Shell Structure $CoFe_2O_4$@Fe_3C@NiO Composite with High Broadband Electromagnetic Wave-Absorption Performance," *Nanoscale*, vol. 15, no. 40, pp. 16381–16389, 2023. doi:10.1039/D3NR03837H.

Conference Proceedings and Presentations

1. X. Xue, **Yu Chu**, S. Moon, N. Sharma, and X. Jiang, "Wearable Ultrasound Transducer Materials and Devices for Applications in Sensing, Imaging, and Therapy," *Biologically Inspired Materials, Processes, and Systems (BIMPS) 2026*, Proc. SPIE 13944, 2026. doi:10.1117/12.3089961.
2. **Yu Chu**, X. Xue, H. Wu, B. Kreager, and X. Jiang, "A Wearable Ultrasonic Cardiovascular-Twin System for Continuous, Non-Invasive Blood Pressure and Hemodynamic Monitoring," oral presentation, 50th Annual Symposium on Tissue Characterization, 2026.
3. X. Xue, **Yu Chu**, S. Moon, V. Ganesh, N. Sharma, and X. Jiang, "Muscle-Resolved Tremor Characterization Using Wearable Ultrasound Tissue Displacement During Task-Evoked Tremor," oral presentation, 50th Annual Symposium on Tissue Characterization, 2026.
4. **Yu Chu**, X. Xue, S. Liu, H. Wu, B. C. Kreager, A. O. Biliroglu, O. Oralkan, and X. Jiang, "Wearable Ultrasound Blood Pressure Monitoring System for Cardiovascular Health," *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, pp. 1–6, 2024. doi:10.1109/UFFC-JS60046.2024.10793899.
5. J. Wang, H. Wu, M. Chen, **Yu Chu**, C. Shi, and X. Jiang, "Stability of Fractional Vortex Ultrasound via a Piezoelectric Transducer," *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, pp. 1–5, 2024. doi:10.1109/UFFC-JS60046.2024.10793625.

Selected Academic Projects

Carnegie Mellon University Pittsburgh, PA
Project Manager, Consulting Project for Dollar Bank January 2023–May 2023

- Collaborated with Dollar Bank's Digital Management Team to identify reliability and process-improvement opportunities for digital banking services.
- Researched European fintech practices and proposed cost-effective strategies for improving retail-banking processes.

Carnegie Mellon University Pittsburgh, PA
Income Prediction and Analysis with Machine Learning January 2022–May 2022

- Implemented SVM, decision-tree, naive Bayes, and logistic-regression classification models using Weka Explorer.
- Improved average model accuracy from 80.82% to 92.69% through feature-space design, error analysis, and hyperparameter tuning.

University of California, Santa Barbara Santa Barbara, CA
Senior Capstone Project with ASML August 2020–June 2021

- Designed and optimized an optical-transmission model with computer-vision algorithms for target-metric measurement.
- Improved thresholding and shape-fitting methods to measure shadow contrast and estimate tin-target morphology.

Leadership and Professional Activities

Chinese Students and Scholars Association, Carnegie Mellon University October 2021–October 2022
Undersecretary, Events Department. Organized student-oriented events and coordinated event planning and public-account articles.

Chinese Students and Scholars Association, University of California, Santa Barbara May 2018–May 2019
Undersecretary, Public Relations Department. Conducted sponsor negotiations, drafted contracts, planned activities, and trained new members.

IEEE Student Activities, University of California, Santa Barbara September 2017–June 2018
Built technical student projects including a Bluetooth-controlled lamp using C and TI LaunchPad and a Python implementation of the classic Snake game.

Technical Skills

Research Methods	Ultrasound transducer design and characterization; wearable ultrasound sensing; pulse-echo signal acquisition; physiological waveform analysis; KLM modeling; COMSOL multiphysics simulations; benchtop, phantom, ex vivo, and in vivo validation.
Programming	Python, MATLAB, C/C++, Java, HTML, Verilog, Unity; machine-learning workflows with PyTorch, TensorFlow, Weka, and LightSide.
Engineering Tools	COMSOL, SolidWorks, AutoCAD, Adams, CoDeSys, DIGSI, LTspice, Arduino, FPGA, Blender, Fritzing, Excel.

Last updated: June 9, 2026.